

## **7.1.4 Software Detailed Design Process**

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### **7.1.4.1 Purpose**

The purpose of the Software Detailed Design Process is to elaborate the software architectural design into a detailed design of software components and units that can be implemented, verified, and integrated. This process refines architectural elements into software units suitable for coding, testing, and integration.

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### **7.1.4.2 Outcomes**

As a result of successful implementation of the Software Detailed Design Process:

- a) A detailed design of each software component is developed, describing the software units to be built.
  - b) External interfaces of each software unit are defined.
  - c) Consistency and traceability are established between the detailed design, software requirements, and architectural design.
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### **7.1.4.3 Activities and Tasks**

The project shall implement the following activities in accordance with applicable organizational policies and procedures.

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#### **7.1.4.3.1 Detailed Design Development**

This activity develops the detailed design of software components, interfaces, databases, documentation, and testing plans.

- **7.1.4.3.1.1** Develop a detailed design for each software component of the software item. Refine components into lower levels containing software units that can be coded, compiled, and tested. Ensure all software requirements are allocated from components to units. Document the detailed design.
- **7.1.4.3.1.2** Develop and document detailed designs for interfaces external to the software item, between software components, and between software units. The designs shall be detailed enough to permit coding without further information.

- **7.1.4.3.1.3** Develop and document a detailed design for the database associated with the software item.
  - **7.1.4.3.1.4** Update user documentation as necessary to reflect detailed design decisions.
  - **7.1.4.3.1.5** Define and document test requirements and schedules for testing software units, including stressing the software unit at its operational limits.
  - **7.1.4.3.1.6** Update the test requirements and schedule for Software Integration to reflect detailed design decisions.
  - **7.1.4.3.1.7** Record detailed design artifacts in accordance with the **Software Documentation Management Process** (7.2.1) and place them under configuration control in accordance with the **Software Configuration Management Process** (7.2.2).
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#### **7.1.4.3.2 Evaluation and Review of Detailed Design**

This activity evaluates the detailed design to ensure its quality, feasibility, and alignment with requirements and architecture.

- **7.1.4.3.2.1** Evaluate the software design and test requirements against the following criteria and document the results:
    - a) Traceability to the software item's requirements.
    - b) External consistency with the architectural design.
    - c) Internal consistency between software components and units.
    - d) Appropriateness of design methods and standards.
    - e) Feasibility of testing.
    - f) Feasibility of operation and maintenance.
  - **7.1.4.3.2.2** Conduct formal review(s) of the software detailed design in accordance with the **Software Review Process** (7.2.6).
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#### **Cross-References**

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for allocation and traceability of requirements to units.
- **Software Architectural Design Process (7.1.3)** — for consistency with architectural structures.
- **Software Documentation Management Process (7.2.1)** — for recording design artifacts.
- **Software Configuration Management Process (7.2.2)** — for baselining detailed design.

- **Software Review Process (7.2.6)** — for conducting detailed design reviews.